Robotic Manipulation

Exercise 6 - Report

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• My choice of matrix P

My chosen selection/choice matrix for controlling force along x, and controlling position along y and z:

$$P = \begin{bmatrix} \mathbf{0} & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathbf{1} \end{bmatrix}$$

I relaxed the position controlling by putting 1's to the moment axes too. With that, I allow rotation around x, y and z axes, in order to enable the robot to reach the desired positions.

• Mathematical equations I used to compute position error

The position error was calculated with the following line of code:

```
err = desired_pose.linear() * (desired_pose.inverse() * current_pose).translation();
```

where the *desired*_pose is the pose we want our robot to have, and the *current_pose* is where our robot currently is.

In our case the *linear()* function returns the rotational part of the transformation matrix.

<u>The mathematical equation: $error = desired_pose^{R} * (desired_pose^{-1} * current_pose)^{T}$</u>

The *desired_pose* is calculated with the following line:

```
Eigen::Isometry3d desired_pose = Eigen::Translation3d(0.0, y_coord, z_coord) * starting_pose;
```

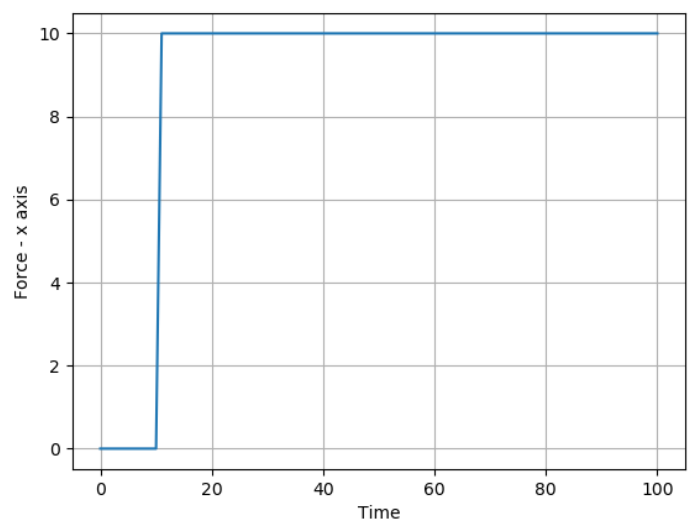
where the desired translation is multiplied with the initial position in order to get the desired pose.

<u>The mathematical equation: *desired_pose = desired_translation * starting_pose*</u>

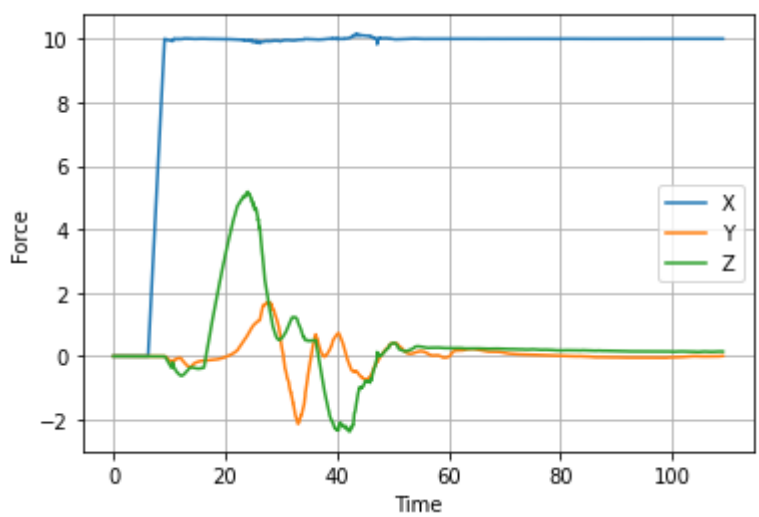
The *desired_translation*'s y and z coordinates are computed as lines, in order to follow the desired path given in the documentation. For example between 10 and 20 seconds the y and z coordinates are computed in the following way:

```
z_coord = (0.02 * time_from_start) - 0.2;
y_coord = 0;
```

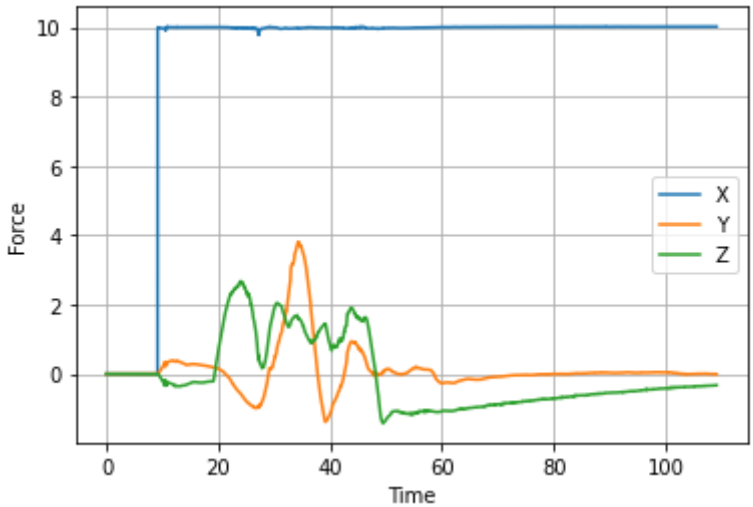
• Plots of the desired and applied force



The desired force along x axis on robot1

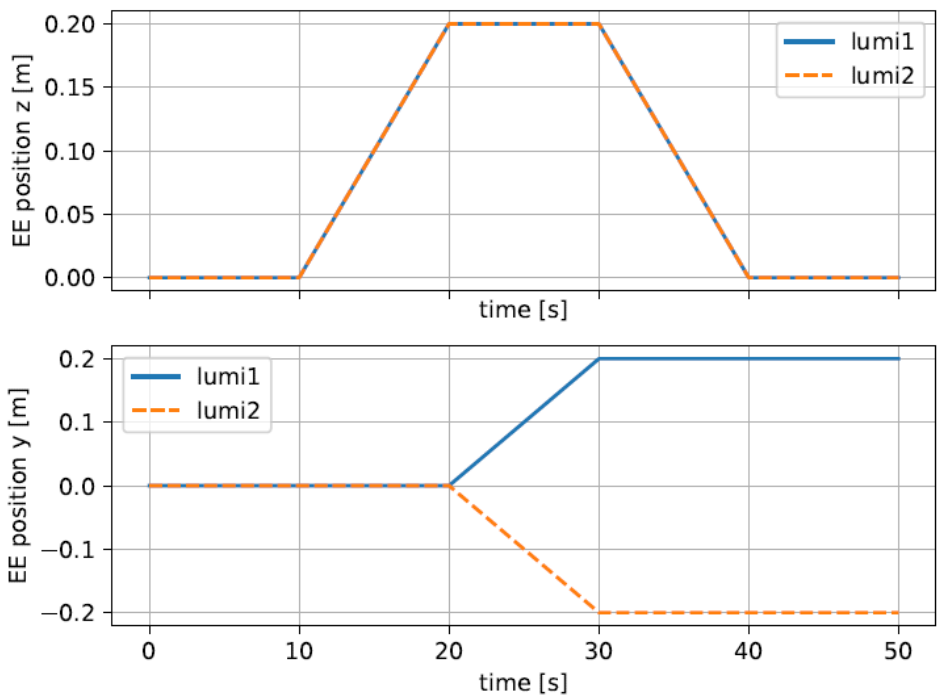


The applied forces on robot1

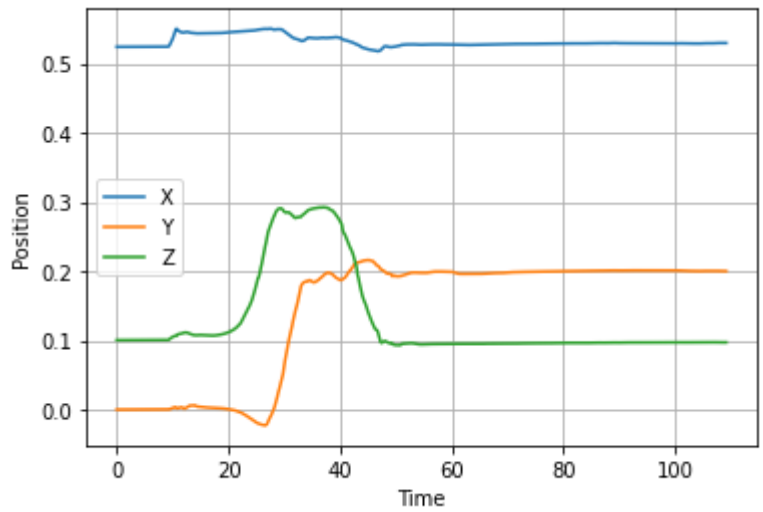


The applied forces on robot2

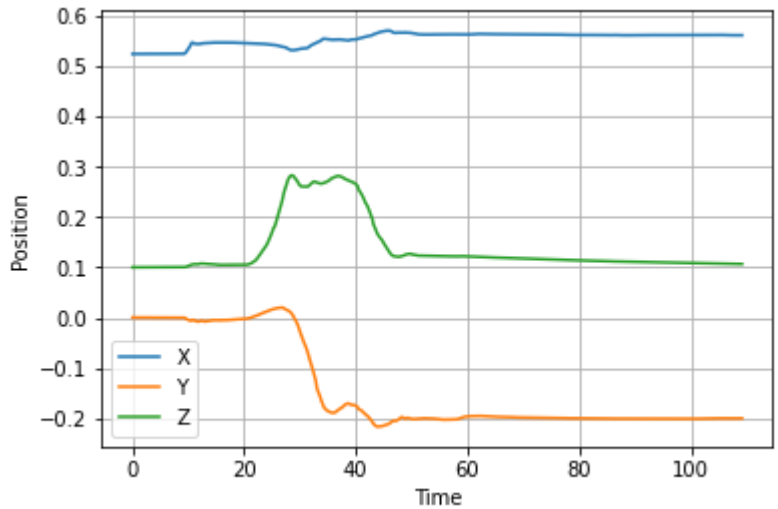
• Plots of the desired and followed paths



The desired paths of robot1 and robot2



The followed paths of robot1



The followed paths of robot2

• Discussion of the results and answers to the following questions:

- **Is the contact force equal to the desired force?**

  One of our goals was to control the contact forces along the *x* axis. As we can see on the graphs, for both *robot1* and *robot2*, the contact force along *x* quickly meets the criteria that we set as desired value:

$$\boldsymbol{f}_d = \begin{pmatrix} 10 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}^\top$$

  Since we only want to control the contact forces along *x*, we don't care about the contact forces along *y* and *z*.

- **Does the followed trajectory correspond to the desired one?**

  Our other goal was to control the position along *y* and *z*. It is discernible on the graphs that the robots follow the desired trajectory.

- **Are all cartesian degrees of freedom followed equally accurately? If not, explain why.**

  The trajectory is more or less accurate along the *y* axis, with a little overshoot at both edges. However, the position control along the *z* axis is less accurate. It is because it has more disturbing factors, such as ***gravity***.

• Estimate of time spent on this exercise.

14 hours